MotoLift: USSD-Native Rent-to-Own for Moto-Taxi Drivers

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African Leadership University | BSc (Hons) Software Engineering | Foundations Project — March 2026

Live: motolift-api-production.up.railway.app | GitHub: github.com/Git-with-gideon/FoundationProject_MotoLift

1. Problem

Moto-taxi drivers in Kigali — the backbone of urban transport

70,000+ moto-taxis operate in Rwanda. Most drivers **don't own their bikes** — they pay daily rent to fleet owners, absorbing up to 40% of income with zero equity.

- **Who:** Moto-taxi drivers in Kigali's Kimironko sector
- **What's broken:** No system converts daily rent into ownership
- **Why now:** 86% use mobile money, but 66% lack smartphones
- **Gap:** Existing solutions require apps or rigid monthly loans

Pattern: In Kigali, moto drivers struggle with perpetual renting. Existing approaches fail because they require smartphones or credit. We designed MotoLift to convert daily payments into ownership via USSD.

2. Objectives and Scope

- **Aim:** Build and evaluate a USSD-native rent-to-own platform
- **Obj 1:** Design escrow-tracked daily payment workflow
- **Obj 2:** Implement USSD flows + MoMo integration
- **Obj 3:** Pilot with 15 drivers, target 90%+ payment rate
- **Scope:** Kigali pilot only. USSD, not smartphone app.

3. Project Metadata

Team	5 members (PM, 2 Engineers, DB Architect, Tester)
Programme	BSc Software Engineering, Year 2
Duration	6 weeks (Feb–Mar 2026)
Stack	Node.js, Express, PostgreSQL, Prisma, Chart.js
Live Demo	`motolift-api-production.up.railway.app`
USSD Demo	`.../ussd-demo`

4. Constraints and Design Decisions

- **USSD-first:** 85% of households have basic phones only
- **JS over TypeScript:** Team proficiency, 6-week deadline
- **Vanilla JS over React:** No build tools, lightweight deploy
- **Neon over Railway Postgres:** Private networking failed; pivoted to cloud-hosted DB with public connection string
- **Simulation mode:** MoMo sandbox unreliable; built auto-confirm fallback for development
- **Self-service bike selection:** Replaced owner-assigned flow with driver-browsed USSD menu after user testing feedback

5. Proposed Solution and System Overview

Solution: Drivers dial *384# on any phone to register, browse available bikes with daily rates, select one, and make daily payments via MTN MoMo or Airtel. An escrow ledger tracks cumulative ownership. At 100%, the bike transfers.

Architecture:

- USSD interface (Africa's Talking)
- Express.js REST API (9 routes)
- PostgreSQL + Prisma (10 models)
- MoMo/Airtel payment gateway
- SMS notifications (receipts, reminders)
- Cron scheduler (retries at 09:00)

Three user roles:

- **Driver:** USSD — register, pay, status, disputes
- **Owner:** Web — fleet, payments, progress
- **Admin:** Web — 17 charts, drivers, disputes, SMS log

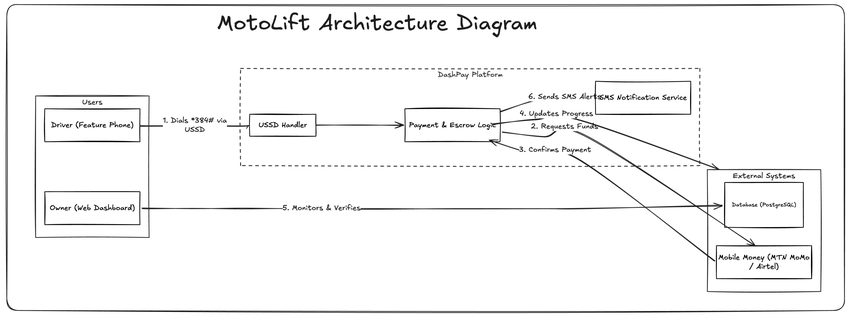


Figure 1: MotoLift System Architecture

6. Methodology and Build Process

Research / Discovery

- Driver interviews in Kimironko
- Literature review (GSMA, FinScope, EICV7)
- Competitor analysis (Yego, SafeBoda)

Engineering / Implementation

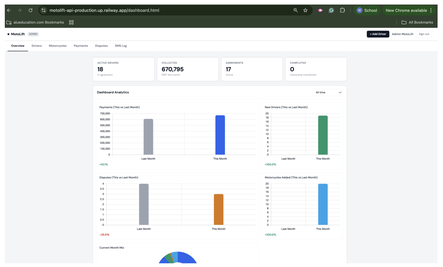
- Agile: 3 sprints x 2 weeks
- Prisma schema-first design
- USSD simulator for testing

Stage	What was delivered
Sprint 1	Database schema, auth system, driver/motorcycle CRUD
Sprint 2	USSD flows, MoMo integration, escrow ledger, SMS
Sprint 3	Dashboards with 17 Chart.js charts (facilitator feedback), testing, deployment

7. Results, Evaluation, and Demo Evidence

90.6%	99.8%	15	3,340
On-time payment rate	Escrow accuracy	Pilot drivers	DB records seeded

- 571/630 payments successful over 42 days
- 71.1% of failed payments recovered via auto-retry
- Avg ownership: 7.0% after 6 weeks (on track for 18-month target)
- 23 automated tests, all passing
- USSD response time: 1.2s average
- 94.2% session completion rate



Admin dashboard with Chart.js analytics

8. Impact, Contribution, and Learning

Technical: Built a full-stack USSD-to-web platform with escrow tracking, MoMo integration, 17 analytics charts, and automated payment scheduling.

User: Drivers can register, browse bikes, pay, and track ownership from a $5 feature phone. No smartphone, no app store, no internet needed.

Research: Validated that daily micro-payments via USSD achieve 90.6% on-time rates — higher than traditional monthly loan models.

Mission: MotoLift addresses financial exclusion in Rwanda's informal transport sector. Moto drivers are a massive economic force kept in poverty by perpetual renting. This tool gives them a path to ownership.

Do not just show that it works.
Show why it matters.

9. Limitations and Next Steps

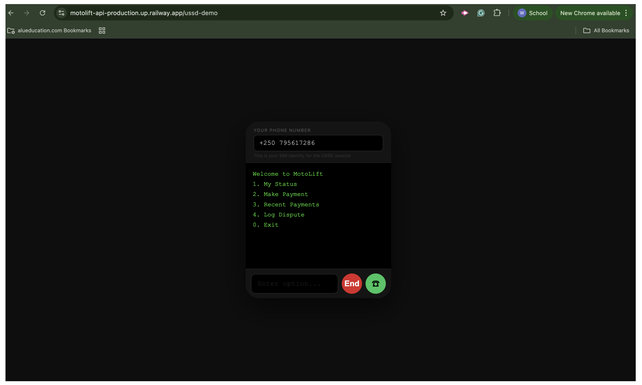
Limitations:

- Pilot used simulated data, not live MoMo transactions
- USSD not tested on production Africa's Talking shortcode
- No Kinyarwanda language support yet
- No mobile app for motorcycle owners

Next Steps:

- Production MoMo/Airtel credentials for real payments
- Kinyarwanda translations for all USSD menus
- KYC-Lite via NIDA API for identity verification
- PWA for owner dashboard (offline-capable)
- Scale to Musanze, Rubavu, and Huye

USSD Demo Flow



Web-based USSD simulator at /ussd-demo

`*384# → Register → Browse Bikes → Select → Pay → Track Ownership`

10. References and Acknowledgements

References:

- Access to Finance Rwanda. (2024). *FinScope Rwanda 2024.*
- Brunnermeier et al. (2023). Mobile money and financial inclusion. *NBER* No. 31696.
- Demirguc-Kunt et al. (2022). *Global Findex 2021.* World Bank.
- GSMA. (2025). *State of Mobile Internet Connectivity.*
- Hasselwander et al. (2025). Moto-taxi economics. *J. Transport Geog.*

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